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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Academic Year: 2025-26	Mini Project-II Synopsis		Sem-VI
	Department : Computer Science and Engineering		Date of Preparation: 16/03/2026
Roll No	3029,3030,3031,3032,3043	Class	T.Y.CSE
Domain Name	Web Development, Environmental Technology		
Project Title	EcoCarbonX		
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Introduction:

Climate change is one of the most critical global challenges caused by increasing greenhouse gas emissions, especially carbon dioxide (CO_2). To control emissions, the concept of **carbon credits** has been introduced, where one carbon credit represents the reduction of one ton of CO_2.
Carbon trading allows organizations to buy and sell these credits, encouraging environmentally friendly practices. However, in India, the carbon credit ecosystem is still developing and lacks a unified, transparent, and user-friendly digital platform.
The proposed system, **EcoCarbonX**, aims to simulate a carbon credit marketplace where users can register eco-friendly projects, generate carbon credits, and trade them. The system will also include mechanisms for credit tracking and retirement to ensure transparency and avoid misuse.

Literature Review :

Carbon credit trading in India has evolved significantly over the past two decades, influenced by global climate agreements such as the domestic policy initiatives. Early studies highlight that India has been an active participant in the **Clean Development Mechanism (CDM)**, contributing substantially to global carbon markets through projects in renewable energy, agriculture, and waste management. Between 2010 and 2022, India generated a large volume of carbon credits, indicating its potential as a major supplier in international carbon markets ().
Research literature identifies that India initially relied on **market-based mechanisms** such as the Perform, Achieve, and Trade (PAT) scheme and Renewable Energy Certificates (REC), which focus on improving energy efficiency and promoting renewable energy adoption.
Recent studies emphasize the introduction of the **Carbon Credit Trading Scheme (CCTS), 2022**, which provides a legal framework for establishing the **Indian Carbon Market (ICM)**. This framework defines institutional roles, including regulatory bodies and technical committees, and enables the issuance of Carbon Credit Certificates (CCCs) based on emissions reduction

performance. However, literature suggests that the Indian carbon market is still in its early developmental stage and requires further refinement to align with global standards ().
Several research papers highlight key challenges in the Indian carbon credit ecosystem. These include lack of transparency, high transaction and verification costs, and limited participation of small-scale stakeholders such as farmers and small enterprises. Studies also reveal socio-economic issues, where benefits of carbon projects are often not equitably distributed, with marginalized groups having limited participation.

Relevance of the Work:

The proposed system is relevant due to the following reasons:

- Promotes environmental sustainability by encouraging carbon reduction.
- Supports India's transition toward a structured carbon market.
- Enables small contributors (farmers, NGOs) to participate.
- Provides a digital solution to improve transparency and accessibility.

Proposed Work:

1. User Management Module

The platform will support multiple user roles:

- **Project Owners** (farmers, NGOs, companies)
- **Buyers** (organizations or individuals looking to offset emissions)
- **Admin/Verifier** (system authority)

Each user will be able to register, log in, and manage their profile. Role-based access control will ensure that each user interacts only with relevant functionalities.

2. Project Registration and Management

Project owners will be able to submit eco-friendly projects such as:

- Tree plantation
- Solar energy generation
- Wind energy projects

The submission will include detailed inputs like location, project size, expected carbon reduction, and supporting evidence (images/documents).

3. Verification and Approval System

An admin panel will simulate the role of verification authorities. Admins will:

- Review submitted projects
- Approve or reject based on provided data
- Assign project status (Pending / Approved / Rejected)

4. Carbon Credit Generation Engine

Once a project is approved, the system will generate carbon credits using predefined estimation models. For example:

- Tree-based projects → credits based on number of trees and absorption rate
- Solar projects → credits based on energy generated and emission factor

Each generated credit will include:

- Unique Credit ID
- Linked Project ID
- Quantity (in tons of CO_2)
- Status (Available / Sold / Retired)

5. Marketplace and Trading

The platform will include a marketplace where project owners can list their carbon credits for sale. Features include:

- Price setting per credit
- Available quantity display
- Buyer browsing and filtering

Buyers can purchase credits based on their requirements, simulating a real carbon trading environment.

6. Wallet and Transaction Management

Each user will have a digital wallet to:

- Store purchased or generated credits
- Track balance
- View transaction history

The system will record all transactions, including buying, selling, and transfer of credits, ensuring traceability.

7. Carbon Emission and Offset Module

Buyers will be able to input their emission data (e.g., fuel usage, electricity consumption). The system will:

- Calculate estimated CO_2 emissions
- Suggest required number of carbon credits for offset

This feature enhances usability and provides practical insights into carbon footprint management.

8. Credit Retirement Mechanism

A key feature of the system is the retirement mechanism, which ensures that each carbon credit is used only once. When a buyer offsets emissions:

- Credits are deducted from the wallet
- Status is updated to "Retired"
- Credits cannot be reused or resold

10. Dashboard and Analytics

The platform will provide dashboards for users and admins displaying:

- Total credits generated
- Credits sold and retired
- Total CO_2 offset
- User activity statistics

This helps in monitoring system performance and environmental impact.

Proposed Methodology:

The system will follow these steps:

1. **User Registration:** Users create accounts as project owners or buyers.
2. **Project Submission:** Project owners submit project details.
3. **Verification:** Admin reviews and approves/rejects.
4. **Credit Generation:** Credits are generated.
5. **Marketplace Listing:** Credits listed for sale.
6. **Purchase:** Buyers purchase credits.
7. **Retirement:** Credits marked as retired.

HW/SW Requirement:

Hardware Requirements

- System with minimum 4 GB RAM
- Processor: Intel i3 or above
- Internet connection

Software Requirements

- Frontend: HTML, CSS, JavaScript
- Backend: Python,
- Database: PostgreSQL / MongoDB
- Tools: VS Code, GitHub
- Browser: Chrome / Edge

Flowchart/Algorithm:

System Architecture/ Flowchart/Algorithm Diagrams……

References:

1.International Carbon Action Partnership : Indian Carbon Credit Trading Scheme | International Carbon Action Partnership

2.Press Information Bureau: Press Note Details: Press Information Bureau

3.ChatGPT : ChatGPT

Signature of Student

1. Satyajeet.D.More –
2. Onkar.K.Patil –
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5. Sadabashya.A.Mujawar –

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